

Report: Twitter Sentiment Analysis Web App using Streamlit and BERT

1. Objective

The goal of this project is to create an interactive **Twitter Sentiment Analysis** web application using a **BERT-based classification model**. The application is deployed using **Streamlit** and made publicly accessible via **ngrok**, making it suitable for testing and demonstration in environments like Google Colab.

2. Main Components

a. Package Installation

```
!pip install streamlit pyngrok transformers torch
```

Installs essential packages:

- **streamlit**: For building the web UI
- **pyngrok**: To expose the local app to the internet
- **transformers** & **torch**: For loading and running the BERT model

b. Colab Keep-Alive Script

A JavaScript snippet is injected to keep the Colab notebook session alive by sending periodic pings.

c. Streamlit App (app.py)

The app is written to a file named **app.py** with the following functionalities:

- Loads a fine-tuned BERT model from a local directory **sentiment_model**
- Defines a **predict()** function to classify input text as **Positive**, **Neutral**, or **Negative**
- Implements a clean UI with input, examples, and result feedback using emojis and styling

d. Model Caching

```
@st.cache_resource
def load_model():
```

This function caches the model and tokenizer to improve performance and avoid reloading them on every interaction.

e. App Deployment with ngrok

The **run_streamlit_with_ngrok()** function:

- Terminates existing Streamlit processes
- Starts a new Streamlit session on port **8501**
- Authenticates and connects ngrok to expose the app publicly

3. Output

After execution, a public URL [via ngrok] is generated where users can:

- Input any tweet-like text
- Analyze its sentiment
- Test pre-filled example inputs

4. Strengths

- Easy deployment in Colab

- Interactive UI with examples
- Efficient model loading via caching
- Real-time sentiment classification using a powerful NLP model (BERT)

5. Areas for Improvement

- Include model training/loading explanation or fallback if **sentiment_model** is missing
- Use environment variables or safer practices for ngrok token
- Add logging or exception handling for robustness

6. Conclusion

This project is a simple yet effective demonstration of deploying NLP models with Streamlit. It provides a user-friendly interface for real-time sentiment analysis and showcases how to integrate ngrok for app sharing without complex backend setups.